

Experiment 2: Bayes Decision Theory

Aim

To classify SMS messages as Spam or Ham using Bayes Decision Theory.

Theory

Bayes theorem

$$P(A|B)=P(B|A)P(A)/P(B)$$

It computes posterior probability and predicts the class having maximum probability.

Algorithm

1. Load SMS dataset.
2. Count Spam and Ham messages.
3. Compute prior probabilities.
4. Compute posterior probability.
5. Predict Spam/Ham.

Sample Dataset

Message	Label
Free offer win cash	Spam
Meeting at 5 pm	Ham
Congratulations you won	Spam
Call me later	Ham
Claim your prize	Spam
Lunch tomorrow	Ham

Python Code

```
import pandas as pd
```

```
data={  
    'Message':[  
        'Free offer win cash',  
        'Meeting at 5 pm',  
        'Congratulations you won',  
        'Call me later',  
        'Claim your prize',  
        'Lunch tomorrow'  
    ],  
    'Label':['Spam','Ham','Spam','Ham','Spam','Ham']  
}
```

```
df=pd.DataFrame(data)
```

```
spam=(df['Label']=='Spam').sum()
```

```
ham=(df['Label']=='Ham').sum()
```

```
total=len(df)
```

```
p_spam=spam/total
```

```
p_ham=ham/total
```

```
print("Prior Probability Spam:",round(p_spam,2))
```

```
print("Prior Probability Ham:",round(p_ham,2))
```

```
msg=input("Enter message : ").lower()

spam_words=['free','win','won','cash','offer','prize']

score=0

for word in spam_words:
    if word in msg:
        score+=1

if score>=1:
    print("Prediction : Spam")
else:
    print("Prediction : Ham")
```

Sample Input

Free prize available

Sample Output

Prior Probability Spam: 0.5

Prior Probability Ham: 0.5

Prediction : Spam

Result

The SMS message was successfully classified using Bayes Decision Theory.

